

DAT 5564: Database Design and SQL

Fall 2025, Mini A

Lab #2: SQL Part 1

INSTRUCTIONS

1. This is a group assignment to be completed during the lab.
2. Please include only your Student IDs on the submission.
 - a. This submission will be submitted online on Canvas.
 - b. Each member of your group needs to submit, even though you are working in a group.
 - c. This submission is automatically graded.
 - d. The submission is due on Canvas.

SESSION OUTLINE

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|------------------------------------|---------------|
| • Introduction of the database | 5 minutes |
| • Lecture: Joining Tables (Part 1) | 10 minutes |
| • Assignment Questions #1 – #13 | 40 minutes |
| • Solutions to some questions | 10-15 minutes |

ASSIGNMENT

Lyft is a well-known ride service that allows individuals to drive for Lyft. In essence, Lyft replaces traditional taxi service. Customers who want to use the service request a ride using the Lyft app. Using their proprietary technology, Lyft finds the closest driver to the customer and assigns the driver to the requested ride.

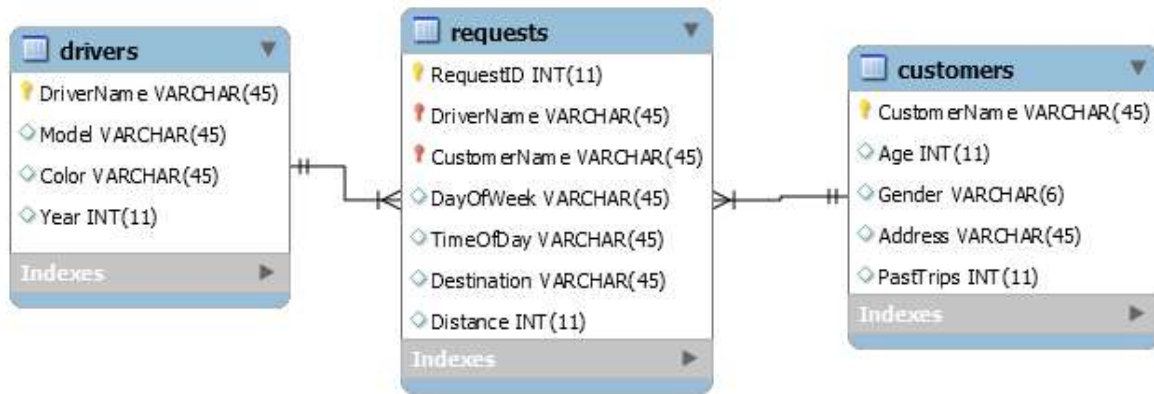
Understanding customer trends is essential for Lyft. The company wants to know its customers' preferences for rides, including when customers are more likely to request a ride and when drivers are available.

Using a sample of ride requests from one week, Lyft compiled data for you to analyze. For this analysis, assume that each ride costs \$10.

Use the Lyft database to answer all questions in this assignment.

The database is posted on Canvas

The E/R diagram for this database is below:



- 1) For each driver, find the driver's name, car color, and car year
- 2) Find all requests for rides on Friday. Identify the driver and the customer. Sort the results by customer name.
- 3) Find all requests for rides on evenings on the weekends (Saturday / Sunday). Identify the destination. Sort results by RequestID.
- 4) Calculate the number of requests and total distance driven by time of day on each day
- 5) Find the average year of manufacturing for each car color. Sort the results in decreasing order of average manufacturing year.
- 6) Create a list of destinations that customers chose this week. Include every destination only once. Do not include the NULL destinations. Sort results in decreasing order.
- 7) Develop one query that will help find how many customers do not have addresses.
- 8) Calculate the number of requests for each day of the week
- 9) Explain some of the previous results (in Question #8). Specifically, explain why there might be variation across days.
- 10) For each age level of customers between 30 – 35, find the total number of past trips and the minimum number of past trips for that age group. Also, include the number of customers in that age group. Sort the results by the age of customers. In other words, find the number of trips, etc, for 30-year-olds, 31-year-olds, ... 35-year-olds.
- 11) For each driver of a white car, find all trips. Include the Destination and 2 other attributes in your results. Sort results in decreasing order of customer name
- 12) Find the name, gender, and address of customers who drove to the Stadium this past week
- 13) List the number of requests for each customer. Sort the results in increasing order of the number of requests. Note, include also the customer address.